QUADRILATERALS

Tiling the Plane

✓ **Quadrilateral and tiling:**

- **Quadrilateral (4-gon):** a closed plane figure with four straight sides.
- **Tiling** (covering a surface with copies of shapes without gaps or overlaps) **the plane:** covering the whole plane with copies of a given shape (or shapes) with no gaps and no overlaps. Example: a floor covered with square tiles.

✓ **Tiling with a hinged grid of sticks:**

- A rectangle tiling can be made with two sets of long wooden sticks, one set horizontal and the other vertical, pinned at every crossing point so that they can turn.
- Hold the bottom stick fixed and push the vertical side. Each set of sticks stays parallel, but the angle between the two sets is no longer 90°.
- The rectangles turn into equal **parallelograms**. So the plane can be tiled with copies of **any parallelogram**.

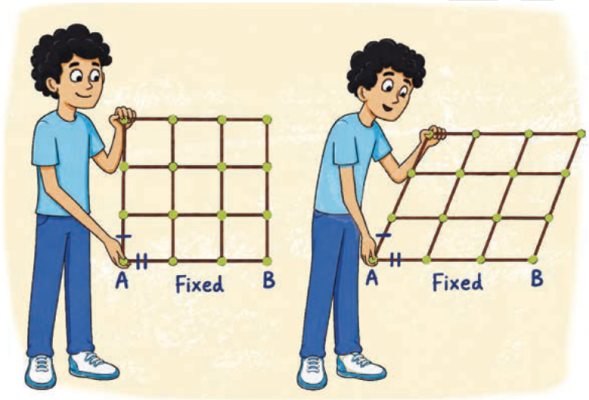


Fig. 1: (A) Grid of rectangles; (B) the same grid pushed into parallelograms

✓ **Solved questions:**

Q1. Can we use any given quadrilateral to tile the plane? If not, which quadrilaterals can be used and which cannot?

➢ **Yes, every quadrilateral can tile the plane**, even an irregular or a non-convex one.

➢ Squares, rectangles and parallelograms tile easily, as the hinged grid shows.

➢ For any other quadrilateral, the four angles add up to 360°, so four copies can be fitted around a point with each angle used once.

➢ Turning copies by 180° about the midpoints of the sides then fills the whole plane (methods in 12.4).

What Exactly is a Quadrilateral?

✓ **Triangle from three points:**

- For three points A, B, C, the triangle △ABC is made of all the points on the segments AB, BC and CA.
- This works only if A, B, C are **not collinear** (lying on one straight line). If they lie on one line, the three segments give just one line segment, not a triangle.
- **Vertices** of △ABC: A, B, C. **Sides**: AB, BC, CA.

Conceptual Learning

✓ **Figures with four points that are not quadrilaterals:**

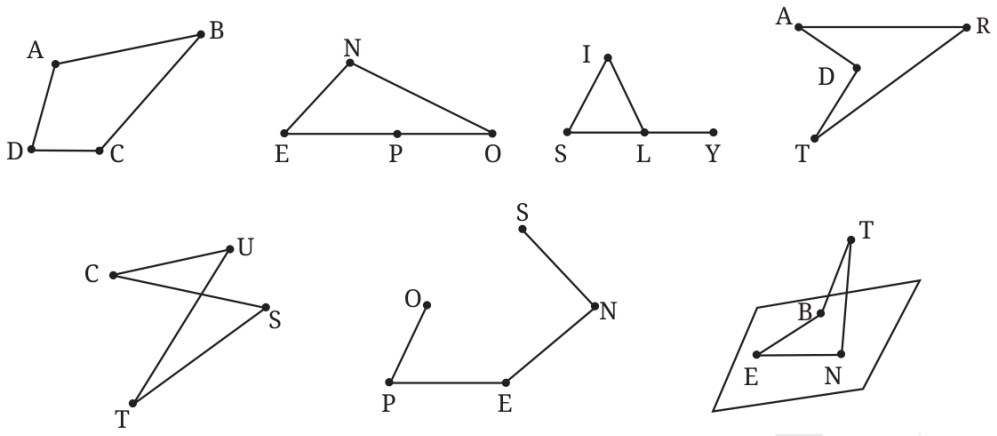


Fig. 2: ABCD, NOPE, SILY, DART (top); CUTS, OPENS, BENT (bottom)

- **ABCD:** an ordinary quadrilateral.
- **NOPE:** E, P, O are collinear, so sides EP and PO lie along one line. The figure is really the triangle NOE.
- **SILY:** S, L, Y are collinear, so sides overlap (YS runs over SL). Not a quadrilateral.
- **DART:** has a dent (an inward bend or hollow) at D, but it is a quadrilateral (a non-convex one).
- **CUTS:** sides CU and TS cross each other. This is a **self-intersecting** 4-gon, not counted as an ordinary quadrilateral.
- **OPENS:** an open chain of segments that does not close up, so it is not a quadrilateral.
- **BENT:** its four vertices are not in one plane. This is a **non-planar (skew)** 4-gon.

✓ **Conditions for a quadrilateral:**

1) **No three vertices collinear** (rules out NOPE and SILY).
2) **All four vertices in one plane** (rules out BENT).
3) **Sides do not cross or touch** except at the vertices (rules out CUTS and OPENS).

- **Quadrilateral ABCD:** for four distinct points A, B, C, D in a plane, the points on the segments AB, BC, CD and DA form quadrilateral ABCD if every such point other than A, B, C, D lies on **exactly one** of these four segments.
- In what follows, "quadrilateral" always means a planar, non-self-intersecting quadrilateral, unless stated otherwise.

✓ **Parts of a quadrilateral:**

- **Vertices:** A, B, C, D. **Sides (edges):** AB, BC, CD, DA. **Diagonals:** AC and BD.
- **Adjacent** (next to each other, sharing a common point) **vertices:** the two endpoints of a side. **Adjacent sides:** two sides with a common endpoint.
- **Internal angles:** the 4 angles, each formed at a vertex by a pair of adjacent sides. **Adjacent angles:** internal angles at adjacent vertices.
- **Opposite sides:** two sides with no common endpoint. **Opposite angles:** angles at two vertices that are not joined by a side.

✓ **Naming a quadrilateral:**

- Quadrilateral ABCD can also be named BCDA, CDAB, DABC, DCBA, ADCB, BADC or CBAD: start at any vertex and go round in either direction (4 × 2 = 8 names).
- Other orders, such as ACBD, join the vertices in a different order and give a different figure.

Conceptual Learning

✓ **Convex and non-convex quadrilaterals:**

- **Convex quadrilateral:** one with no dent; all its internal angles are **less than 180°**.
- **Non-convex quadrilateral:** one internal angle is **more than 180°** (a reflex angle at the dent), like DART.
- **Diagonal test:** the diagonals of a convex quadrilateral cross each other; those of a non-convex quadrilateral do not.

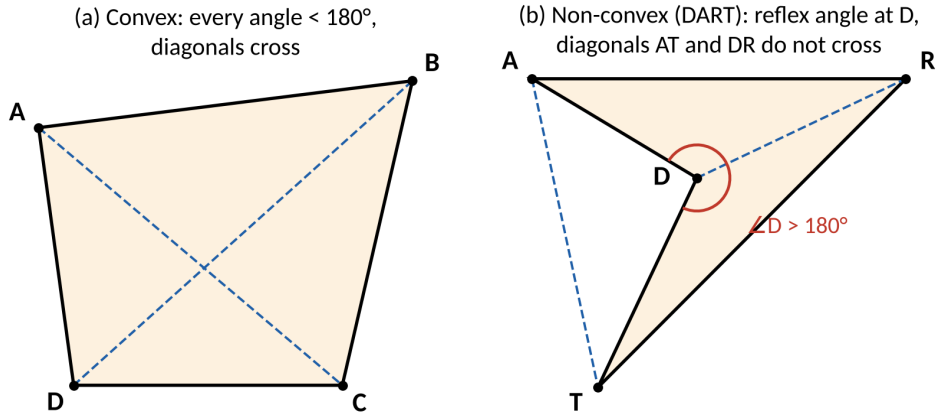


Fig. 3: Convex and non-convex quadrilaterals

✓ **Types of 4-gons:**

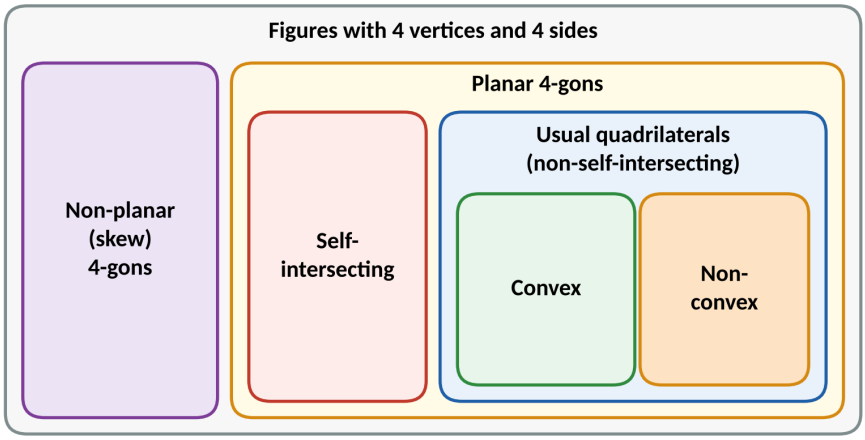


Fig. 4: Types of figures with 4 vertices and 4 sides

✓ **4-gons in machines: linkages:**

- **Linkages** (devices made of bars joined so that they can turn) are machines made of bars joined at their ends. Some ends are fixed, others turn about them.
- A 4-bar linkage can take the shape of a convex, a non-convex or even a self-intersecting 4-gon as it moves. So engineers must study all these shapes.



Fig. 5: A 4-bar linkage in three positions: convex, self-intersecting and non-convex

✓ **Solved questions:**

Q2. Informally, a quadrilateral is a figure with four straight sides, as in the first figure ABCD. But consider the other six figures: the five plane figures NOPE, SILY, DART, CUTS, OPENS and the non-planar BENT. Should we call all these figures quadrilaterals? If you answer 'no' for any of them, how will you describe a quadrilateral so that such a figure is excluded?

- **NOPE: No.** Three vertices are collinear; it is really a triangle.
- Exclude by: no three vertices collinear.
- **SILY: No.** Sides overlap.
- Excluded by the same condition.
- **BENT: No** (as an ordinary quadrilateral)
- It is not in one plane.
- Exclude by: all four vertices in one plane.
- **CUTS: No** (usually)
- Two sides cross.
- Exclude by: every point other than a vertex lies on exactly one side.
- **OPENS: No.** It is not a closed figure; the same condition, with sides AB, BC, CD, DA joining back to the start, excludes it.
- **DART: Yes.** It is a quadrilateral with a dent (non-convex).

Q3. Can we similarly describe a quadrilateral ABCD as the points on segments AB, BC, CD and DA? Take any 4 distinct non-collinear points A, B, C and D. Will segments AB, BC, CD and DA always form a quadrilateral as we visualise it?

- **Not always.** Unlike a triangle, four segments can give strange figures.
- If three vertices are collinear, sides lie along one line (NOPE, SILY).
- If the points are not in one plane, the figure is non-planar (BENT).
- If two sides cross, the figure is self-intersecting (CUTS).
- So extra conditions are needed: four points in a plane, and every point other than a vertex lying on exactly one side.

Q4. Let ABCD be a quadrilateral. (i) List all sides of ABCD adjacent to side AB. List all sides opposite to AB. (ii) List all angles of ABCD adjacent to ∠A. List all angles opposite to ∠A. (iii) Describe a pair of opposite sides and a pair of opposite angles without using the names of the vertices.

- (i) Adjacent to AB: **BC and DA**. Opposite to AB: **CD**.
- (ii) Adjacent to ∠A: **∠B and ∠D**. Opposite to ∠A: **∠C**.
- (iii) **Opposite sides:** two sides that have no common endpoint.
- **Opposite angles:** two internal angles whose vertices are not the two ends of one side.

Q5. You have used internal angles of quadrilaterals, but they too require an exact description, just like how we gave one for a quadrilateral. Precisely describe the internal angle of a quadrilateral at a given vertex. Your answer should work for a non-convex quadrilateral too. (Hint: use the opposite vertex as well.)

- Take vertex A with neighbours B and D, and opposite vertex C. Let ∠BAD be the angle less than 180° between rays AB and AD.

Conceptual Learning

- If A and C lie on **opposite sides** of the line BD, the internal angle at A is **∠BAD**.
- If A and C lie on the **same side** of the line BD, the internal angle at A is **360° − ∠BAD** (a reflex angle).
- Check with DART: at the dent D, the vertices D and R are on the same side of AT, so the internal angle is 360° − ∠ADT > 180°.
- At A, the vertices A and T are on opposite sides of DR, so the internal angle is ∠DAR.

Q6. In a quadrilateral ABCD, suppose AB ∥ DC. Can ABCD be non-convex? What if we instead assume AB = CD? What if we instead assume ∠A = ∠C?

- **AB ∥ DC: No.** A and B lie on one side of line DC, and C and D lie on one side of line AB. With one pair of parallel sides the figure is a trapezium (a quadrilateral with one pair of parallel sides), whose diagonals always cross, so it is **always convex**.
- **AB = CD: Yes**, it can be non-convex.
- Example: take AB = CD = 5 and push D inwards so that the angle at D becomes more than 180°.
- **∠A = ∠C: Yes**, it can be non-convex.
- Example: an arrowhead (dart) symmetric about BD, with equal small angles at A and C and a reflex (more than 180° and less than 360°) angle at B.

Q7. Consider three non-collinear points A, B, C and draw the lines AB, BC, CA. For every possible location of point D in the plane outside these lines, decide if ABCD is self-intersecting, non-convex, or convex. (Hint: the three lines divide the plane into 7 regions.)

- **D inside △ABC:** D is a dent → **non-convex**.
- **D in the region across side AC** (opposite to B): A, B, C, D go round in order → **convex**.
- **D in the region across side AB** (opposite to C): sides AB and CD cross → **self-intersecting**.
- **D in the region across side BC** (opposite to A): sides BC and DA cross → **self-intersecting**.
- **D in any of the 3 corner regions** (beyond a vertex): that vertex falls inside the triangle formed by the other three → **non-convex**.
- Total: 1 region convex, 2 regions self-intersecting, 4 regions non-convex.

Q8. Can a quadrilateral be both self-intersecting and non-planar?

- **No.** If sides AB and CD cross at a point E, the lines AB and CD meet.
- Two lines that meet lie in one plane, so A, B, C, D are all in that plane.
- So a self-intersecting 4-gon is always planar.

Parallelograms

✓ **Parallelogram:**

- **Parallelogram:** a quadrilateral whose opposite sides are parallel. Rhombuses, rectangles and squares are special parallelograms.

✓ **Theorem 1: Properties of a parallelogram:**

- (a) The opposite sides of a parallelogram are equal.
- (b) The opposite angles of a parallelogram are equal.
- (c) The diagonals of a parallelogram bisect each other.
- **Proof of (a):** AC is a transversal (a line that cuts two or more lines) for AB ∥ DC and for AD ∥ BC. Alternate angles give △ACD ≅ △CAB (ASA). So AB = DC and AD = BC.

Conceptual Learning

- **Proof of (b):** BC ∥ AD with transversal AB, so ∠A + ∠B = 180° (interior angles on the same side). Likewise any two adjacent angles add up to 180°. Then ∠A = 180° − ∠B = ∠C and ∠B = ∠D.
- **Proof of (c):** Let the diagonals meet at E. Using AD = BC and alternate angles, △AED ≅ △CEB (ASA). So EA = EC and EB = ED: E is the midpoint of both diagonals.

(a) Diagonal AC: △ACD ≅ △CAB (ASA) (b) Adjacent angles add up to 180° (c) Diagonals bisect each other at E

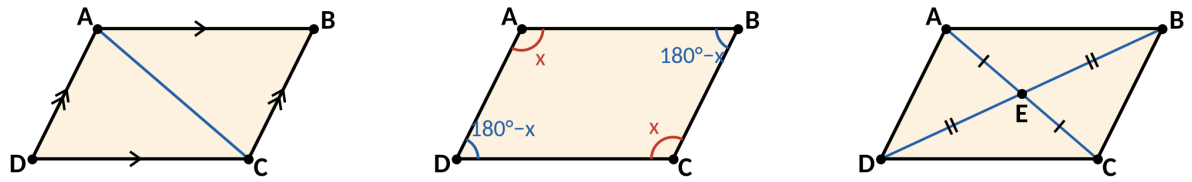


Fig. 6: Proof of Theorem 1 (a), (b), (c)

✓ **Proving converses:**

- A converse (the statement with the given part and the result part swapped) can sometimes be proved by running the original proof **backwards**, giving proper justification for every backward step.
- This does not always work. Example: "If x = y, then $x^2 = y^2$" cannot be reversed, because x = 3, y = −3 gives $x^2 = y^2$ but x ≠ y.

✓ **Theorem 2: Opposite sides equal ⇒ parallelogram:**

1) Let ABCD have AB = DC and AD = BC. Draw diagonal AC.
2) △ACD ≅ △CAB by SSS (AC common).
3) So ∠CAB = ∠ACD and ∠ACB = ∠CAD. These are alternate angles for transversal AC.
4) Hence AB ∥ DC and AD ∥ BC: ABCD is a parallelogram.
5) Same congruence as in Theorem 1(a), but a different test (SSS instead of ASA) and a different result.

✓ **Theorem 3: Opposite angles equal ⇒ parallelogram:**

1) Let ∠A = ∠C = x and ∠B = ∠D = y.
2) Angle sum: 360° = ∠A + ∠B + ∠C + ∠D = 2(x + y), so x + y = 180°.
3) ∠A + ∠B = 180° with transversal AB gives BC ∥ AD.
4) ∠B + ∠C = 180° with transversal BC gives AB ∥ DC. So ABCD is a parallelogram.

- Same result in other words: **if each pair of adjacent angles of a quadrilateral adds up to 180°, it is a parallelogram.**

✓ **Theorem 4: Diagonals bisect each other ⇒ parallelogram:**

1) Let the diagonals meet at E with EA = EC and EB = ED.
2) ∠AED = ∠CEB (vertically opposite), so △AED ≅ △CEB (SAS).
3) So ∠DAE = ∠BCE: alternate angles for transversal AC, hence AD ∥ BC.
4) Likewise △EAB ≅ △ECD gives AB ∥ DC. So ABCD is a parallelogram.

✓ **Theorem 5: Equal and parallel sides test:**

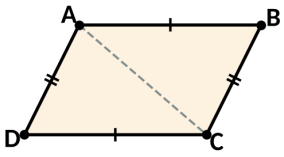
A quadrilateral with one pair of opposite sides equal and parallel is a parallelogram.

1) Let AB ∥ DC and AB = DC. The figure is convex (a trapezium shape), so the diagonals meet at a point E.
2) ∠EAB = ∠ECD and ∠EBA = ∠EDC (alternate angles), with AB = CD, so △EAB ≅ △ECD (ASA).
3) So EA = EC and EB = ED: the diagonals bisect each other.

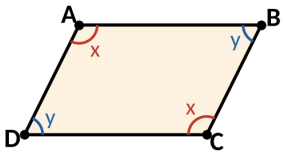
Conceptual Learning

4) By Theorem 4, ABCD is a parallelogram.

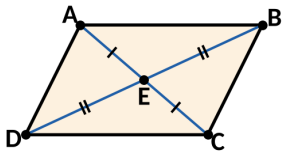
Theorem 2: opposite sides equal



Theorem 3: opposite angles equal
2(x + y) = 360°



Theorem 4: diagonals bisect each other



Theorem 5: AB ∥ DC and AB = DC

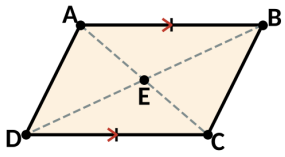


Fig. 7: Tests for a parallelogram: Theorems 2, 3, 4 and 5

✓ **All tests for a parallelogram:**

A quadrilateral is a parallelogram if...	Result
both pairs of opposite sides are parallel	meaning of parallelogram
both pairs of opposite sides are equal	Theorem 2
both pairs of opposite angles are equal (or adjacent angles add up to 180°)	Theorem 3
the diagonals bisect each other	Theorem 4
one pair of opposite sides is equal and parallel	Theorem 5

✓ **Worked examples:**

- **Example 1:** a quadrilateral has sides 5, 7, 5, 7 cm in order. Opposite sides are equal, so it is a parallelogram (Theorem 2).
- **Example 2:** angles 70°, 110°, 70°, 110° in order: opposite angles equal, so it is a parallelogram (Theorem 3). Angles 70°, 110°, 110°, 70° in order: ∠A ≠ ∠C, so it is not a parallelogram.
- **Example 3:** the diagonals of PQRS meet at O with OP = OR = 4 cm and OQ = OS = 6 cm. They bisect each other, so PQRS is a parallelogram (Theorem 4).

✓ **Solved questions:**

Q9. The angles of a quadrilateral add up to 360°. Therefore, if the opposite angles are equal, what can we say about adjacent angles? Is the converse of your answer true? Conclude that Theorem 3 can also be stated as follows. "If each pair of adjacent angles in a quadrilateral ABCD ... then ABCD is a parallelogram." Fill in the blank.

➢ If ∠A = ∠C and ∠B = ∠D, then 2(∠A + ∠B) = 360°, so **adjacent angles add up to 180°**.

➢ Converse: if every pair of adjacent angles adds up to 180°, then ∠A = 180° − ∠B = ∠C and ∠B = 180° − ∠C = ∠D. So opposite angles are equal.

➢ **True.**

➢ Blank: "If each pair of adjacent angles in a quadrilateral ABCD **adds up to 180°**, then ABCD is a parallelogram."

Q10. True or false? (i) A parallelogram with a right angle is a rectangle. (ii) A rhombus with perpendicular diagonals is a square. (iii) If the diagonals of a parallelogram are equal, then it is a rectangle.

➢ (i) **True.** Adjacent angles add up to 180° and opposite angles are equal, so one right angle makes all four angles 90°.

➢ (ii) **False.** The diagonals of every rhombus are perpendicular, so this gives nothing new.

➢ Example: a rhombus with angles 60° and 120° is not a square.

Conceptual Learning

- (iii) **True.** If AC = BD in parallelogram ABCD, then △ABC ≅ △DCB (SSS: AB = DC, BC common, AC = DB)
- So ∠B = ∠C; with ∠B + ∠C = 180°, each is 90°.

Q11. The diagonal AC of a parallelogram ABCD bisects ∠A. Show that it also bisects ∠C and that ABCD is a rhombus.

- ∠BAC = ∠DAC (given).
- AB ∥ DC gives ∠BAC = ∠DCA; AD ∥ BC gives ∠DAC = ∠BCA (alternate angles).
- So ∠DCA = ∠BCA: AC **bisects ∠C**.
- In △ABC, ∠BAC = ∠DAC = ∠BCA, so AB = BC (sides opposite equal angles).
- In a parallelogram AB = DC and BC = AD, so all four sides are equal: ABCD is a **rhombus**.

Q12. The following questions examine converses of true properties. Answer them with Yes or No. If your answer is No, what extra condition can you add so that the answer becomes Yes? (i) If the diagonals of a quadrilateral ABCD bisect its angles, must ABCD be a rhombus? (ii) If the diagonals of a quadrilateral ABCD bisect each other at right angles, must ABCD be a rhombus? (iii) If the diagonals of a quadrilateral ABCD are of equal length, must ABCD be a rectangle? (In the ancient Indian study of quadrilaterals, equality of diagonals was considered significant. Quadrilaterals were first classified according to whether their diagonals were equal or not, before considering equality of sides.)

- (i) **Yes.** AC bisects ∠A and ∠C, so △ABC ≅ △ADC (ASA, AC common): AB = AD and CB = CD. BD bisects ∠B and ∠D, so △BAD ≅ △BCD: AB = CB. All four sides are equal: a rhombus.
- (ii) **Yes.** The diagonals bisect each other, so ABCD is a parallelogram (Theorem 4)
- With E the meeting point, △AEB ≅ △AED (SAS: EB = ED, right angles, AE common), so AB = AD. A parallelogram with two adjacent sides equal is a rhombus.
- (iii) **No.** An isosceles trapezium (or a kite with equal diagonals) has equal diagonals but is not a rectangle.
- Extra condition for (iii): the diagonals also **bisect each other**. Then ABCD is a parallelogram with equal diagonals, which is a rectangle.

Q13. Let ABCD be a parallelogram with AB ≠ BC. Show that the pairwise intersection points of the four angle bisectors form the vertices of a rectangle. Why did we assume AB ≠ BC?

- Let the bisectors of ∠A and ∠D meet at P. ∠A + ∠D = 180° (adjacent angles), so ½∠A + ½∠D = 90°.
- In △APD, ∠APD = 180° − 90° = **90°**.
- In the same way, the bisectors at B and C, at A and B, and at C and D meet at right angles (points R, S, Q).
- So the quadrilateral PQRS has four right angles: it is a **rectangle**.
- If AB = BC, ABCD is a rhombus, its angle bisectors are its diagonals, and all four bisectors pass through one point.
- The rectangle shrinks to a single point.
- So AB ≠ BC is needed.

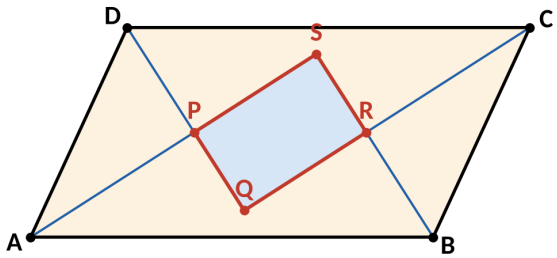


Fig. 8: Angle bisectors of a parallelogram form rectangle PQRS

Conceptual Learning

Q14. In a parallelogram ABCD, two points P and Q are taken on diagonal BD such that DP = BQ. Show that APCQ is a parallelogram.

- Let the diagonals AC and BD meet at O. O is the midpoint of both: OA = OC and OB = OD.
- OP = OD − DP and OQ = OB − BQ. Since OD = OB and DP = BQ, **OP = OQ**.
- The diagonals AC and PQ of APCQ bisect each other at O.
- By Theorem 4, **APCQ is a parallelogram**.

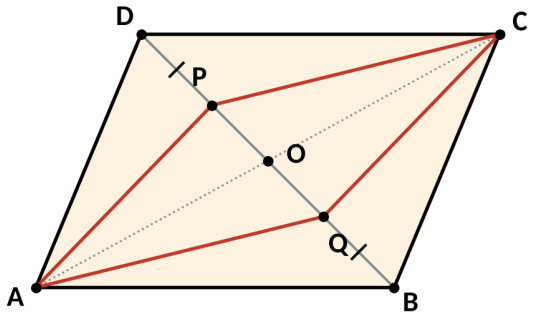


Fig. 9: P and Q on diagonal BD with DP = BQ

Q15. The diagonals AC and BD of a parallelogram ABCD intersect at O. A line through O meets AB and CD at points P and Q respectively. Show that O is the midpoint of PQ. (Multiple proofs are possible. Which is the simplest?)

- OA = OC (diagonals bisect each other).
- ∠OAP = ∠OCQ (alternate angles, AB ∥ DC) and ∠AOP = ∠COQ (vertically opposite).
- So △OAP ≅ △OCQ (ASA), giving OP = OQ. Hence **O is the midpoint of PQ**.
- This congruence proof is the simplest.
- Another way: half-turn about O maps the parallelogram onto itself, AB onto CD, so P goes to Q.

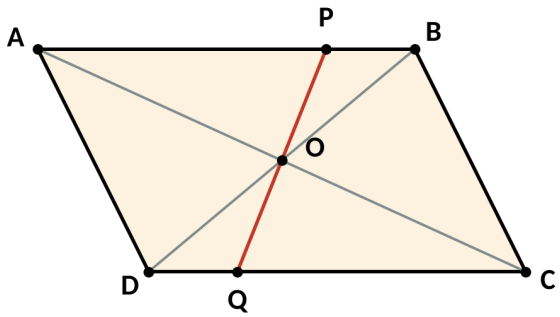


Fig. 10: A line through O meets AB at P and CD at Q

Q16. Review all the properties of a rhombus/rectangle/square that you proved in Grade 8. Formulate a converse of each. Decide if the converse is true. There are many possibilities here!

Property	Converse	True?
Rhombus: diagonals are perpendicular bisectors of each other	If the diagonals of a quadrilateral are perpendicular bisectors of each other, it is a rhombus	Yes
Rhombus: diagonals are perpendicular	If the diagonals are perpendicular, it is a rhombus	No (kite)
Rhombus: diagonals bisect the angles	If the diagonals bisect the angles, it is a rhombus	Yes
Rectangle: diagonals are equal and bisect each other	If the diagonals are equal and bisect each other, it is a rectangle	Yes

Conceptual Learning

Property	Converse	True?
Rectangle: diagonals are equal	If the diagonals are equal, it is a rectangle	No (isosceles trapezium)
Rectangle: all angles are 90°	If all angles are 90°, it is a rectangle	Yes
Square: diagonals are equal, perpendicular and bisect each other	If the diagonals are equal, perpendicular and bisect each other, it is a square	Yes
Square: diagonals are equal and perpendicular	If the diagonals are equal and perpendicular, it is a square	No (a kite can have this)

Applications of Parallelograms

✓ **Uses in machines:**

- A basic law of physics combines forces using parallelograms.
- James Watt's **parallel motion linkage**, part of his steam engine, uses parallelograms.
- A **pantograph** (a jointed device that copies a drawing at a larger or smaller size) is a parallelogram-based device that makes exact scaled copies of a drawing.

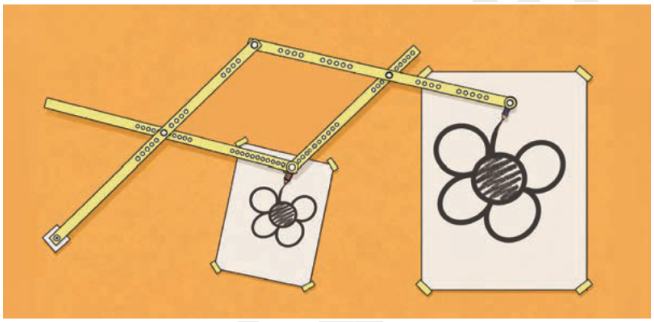


Fig. 11: A pantograph copying a drawing at a larger size

✓ **Four triangles from the midpoints:**

- Join the midpoints of the three sides of a triangle. The triangle is cut into **four congruent** (exactly equal in shape and size) **triangles**, each a smaller copy of the original (proved in the solved questions of this section).
- Proving this directly with congruence tests is hard, as too many lengths are unknown. So we first study **one** segment joining two midpoints.

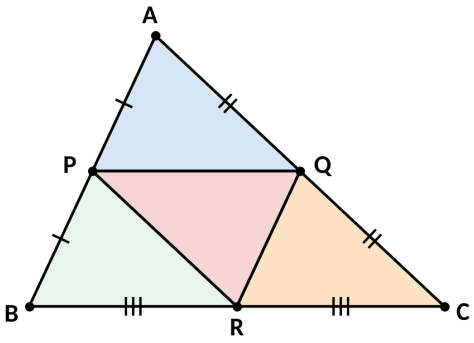


Fig. 12: Midpoints P, Q, R cut △ABC into four congruent triangles

✓ **Theorem 6: Midpoint Theorem:**

The segment joining the midpoints of two sides of a triangle is parallel to the third side and is half of it.

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- **Given:** P and Q are the midpoints of AB and AC. **To prove:** PQ ∥ BC and PQ = BC/2.
- **Construction (key idea):** draw the line l through C parallel to BA, meeting line PQ at R.

1) In △APQ and △CRQ: AQ = CQ, ∠AQP = ∠CQR (vertically opposite), ∠APQ = ∠CRQ (alternate angles, AP ∥ RC).
2) So △APQ ≅ △CRQ (AAS). Hence PQ = QR, so PQ = PR/2, and CR = AP.
3) AP = BP, so CR = BP. Also CR ∥ BP.
4) By Theorem 5, **BCRP is a parallelogram**. So PR ∥ BC and PR = BC.
5) Hence **PQ ∥ BC** and **PQ = PR/2 = BC/2**.

- Such an extra line drawn to help a proof is an **auxiliary** (extra, added to help) **construction**. It is a standard trick learnt with practice.

✓ **Theorem 7: Converse of the Midpoint Theorem:**

The line drawn through the midpoint of one side of a triangle, parallel to another side, bisects the third side (and the segment formed is half the parallel side).

- **Proof 1 (running the Midpoint Theorem proof backwards):**

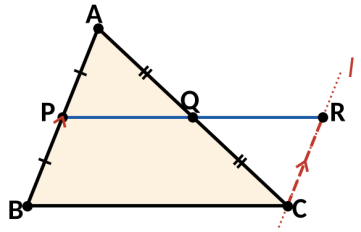
1) P is the midpoint of AB, and the line through P parallel to BC meets AC at Q. Draw the line through C parallel to BA, meeting line PQ at R.
2) BCRP has both pairs of opposite sides parallel, so it is a parallelogram: CR = BP = AP and PR = BC.
3) ∠AQP = ∠CQR (vertically opposite), ∠APQ = ∠CRQ (alternate angles, CR ∥ BA), AP = CR. So △APQ ≅ △CRQ (AAS).
4) Hence **AQ = QC** (Q is the midpoint of AC) and PQ = QR, so **PQ = PR/2 = BC/2**.

- **Proof 2 (using the Midpoint Theorem itself):**

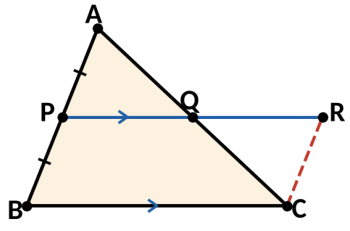
1) Let M be the midpoint of AC. By the Midpoint Theorem, PM ∥ BC.
2) Through P there is only **one** line parallel to BC. So line PM is the same as line PQ.
3) Both lines meet AC at one point, so M = Q. Hence Q is the midpoint of AC.

- Here a theorem is used to prove its own converse; this works because of the extra fact that the parallel line through a point is unique.

(a) Midpoint Theorem: l through C ∥ BA



(b) Converse: PQ ∥ BC, find Q



(c) Second proof: M is the midpoint of AC

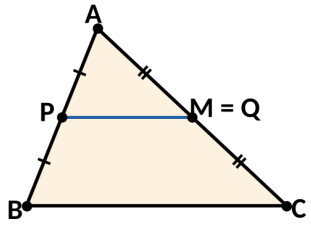


Fig. 13: (a) Midpoint Theorem proof; (b), (c) two proofs of its converse

✓ **Looking ahead: Basic Proportionality Theorem:**

- In higher grades, the condition that P is the midpoint is dropped, keeping only PQ ∥ BC. The result is the **Basic Proportionality Theorem**: AP/PB = AQ/QC.

✓ **Worked examples:**

- **Example 1:** in △ABC, BC = 12 cm. P and Q are the midpoints of AB and AC. Then PQ = **6 cm** and PQ ∥ BC.
- **Example 2:** a triangle has sides 8, 10, 14 cm. The triangle formed by joining the midpoints has sides 4,

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5, 7 cm and perimeter **16 cm** (half of 32 cm).

- **Example 3:** in △ABC, P is the midpoint of AB and PQ ∥ BC meets AC at Q. If AC = 9 cm, then AQ = QC = **4.5 cm** (Theorem 7).

✓ **Medians and the centroid:**

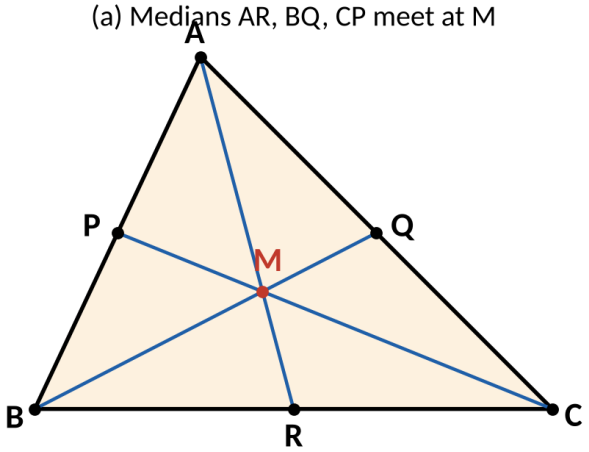
- **Median:** the segment joining a vertex of a triangle to the midpoint of the opposite side. A triangle has three medians.
- **Concurrent** (passing through one common point) **lines:** lines passing through one common point.
- Experiment: the three medians always seem to meet at one point M, which cuts each median in the ratio 2 : 1.

✓ **Theorem 8: Centroid Theorem:**

The three medians of a triangle pass through a common point, which divides each median in the ratio 2 : 1, the longer part being towards the vertex. This point is the centroid.

1) Take medians CP and BQ meeting at M. Let X and Y be the midpoints of BM and CM.
2) Midpoint Theorem in △ABC: PQ ∥ BC and PQ = BC/2. In △MBC: XY ∥ BC and XY = BC/2.
3) So PQ ∥ XY and PQ = XY. With vertically opposite (opposite angles formed where two lines cross) angles at M and alternate angles, △MPQ ≅ △MYX (ASA).
4) So MQ = MX = XB and MP = MY = YC. Hence **CM : MP = 2 : 1** and **BM : MQ = 2 : 1**.
5) Repeat with medians AR and BQ meeting at N: BN : NQ = 2 : 1. Only one point divides BQ in the ratio 2 : 1, so N = M.
6) So all three medians pass through M, which divides each in the ratio 2 : 1.

(a) Medians AR, BQ, CP meet at M



(b) X, Y midpoints of BM, CM: CM : MP = 2 : 1

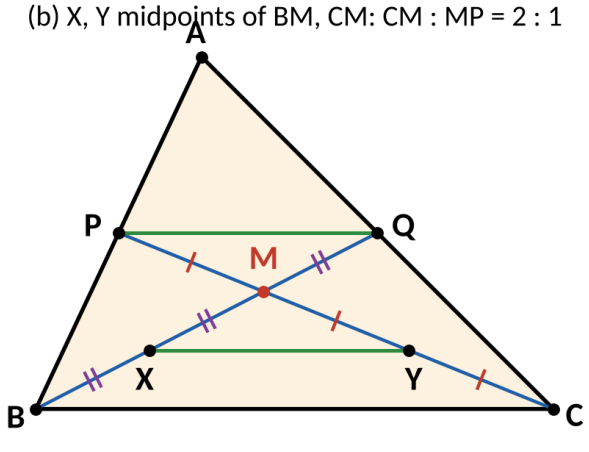


Fig. 14: (a) Medians meet at the centroid M; (b) proof that CM : MP = 2 : 1

✓ **Theorem 9: Midpoint Theorem for Quadrilaterals (Varignon's theorem):**

The midpoints of the four sides of any quadrilateral are the vertices of a parallelogram, called the Varignon parallelogram.

1) Let P, Q, R, S be the midpoints of AB, BC, CD, DA.
2) Midpoint Theorem in △ABC and △ADC: PQ ∥ AC and SR ∥ AC, so **PQ ∥ SR** (also PQ = SR = AC/2).
3) Midpoint Theorem in △BCD and △BAD: QR ∥ BD and PS ∥ BD, so **QR ∥ PS**.
4) Both pairs of opposite sides are parallel, so PQRS is a parallelogram.

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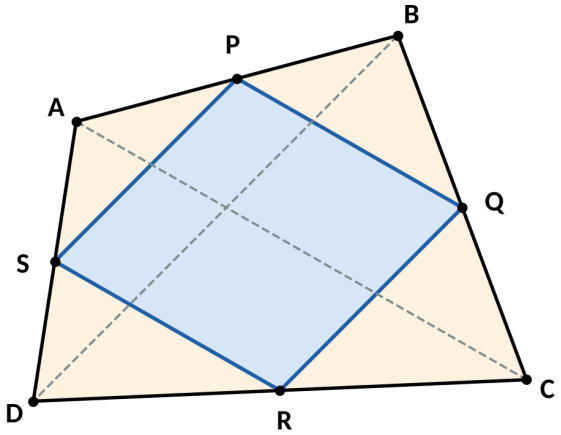


Fig. 15: Varignon parallelogram PQRS of quadrilateral ABCD

- Useful facts: each side of the Varignon parallelogram is half a diagonal of ABCD, so its perimeter = AC + BD.

✓ **Solved questions:**

Q17. Try to prove directly that the four smaller triangles formed by joining the midpoints of the sides of a triangle are congruent.

- A direct attempt fails: in △APQ and △PBR, only AP = PB is known; the other sides and angles are unknown, so no congruence test applies.
- With the Midpoint Theorem it works: PQ = BC/2, QR = AB/2, PR = AC/2.
- So every small triangle has sides AB/2, BC/2, AC/2, and all four are congruent by SSS (full proof in the next question).

Q18. (i) If P, Q, R are the midpoints of sides AB, AC, BC respectively of △ABC, show that △PQR is congruent to △QPA and to two other triangles which you should identify. (ii) Suppose someone erases △ABC, leaving only △PQR on the paper. Can you reconstruct △ABC from △PQR?

- (i) By the Midpoint Theorem: PQ = BC/2 = BR = RC, QR = AB/2 = AP = PB, PR = AC/2 = AQ = QC.
- △PQR and △QPA: PQ common, QR = PA, PR = QA. So △PQR ≅ △QPA (SSS).
- In the same way **△PQR ≅ △RBP** (QR = BP, PQ = RB, PR common) and **△PQR ≅ △CRQ** (PQ = RC, PR = QC, QR common).
- (ii) **Yes.** Through each vertex of △PQR, draw a line parallel to the opposite side: through P parallel to QR, through Q parallel to PR, through R parallel to PQ.
- The three lines meet in pairs at A, B and C. By Theorem 7 (and the uniqueness of parallels), P, Q, R are exactly the midpoints of the sides of this triangle, so it is △ABC.

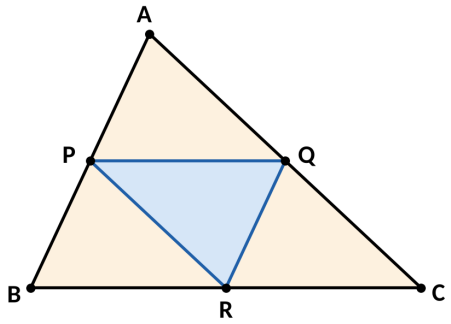


Fig. 16: △PQR formed by the midpoints of △ABC

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Q19. Give another proof of the Centroid Theorem using a clever construction. Let M be the intersection point of the two medians CP and BQ. Show that the line AM meets side BC at its midpoint. For this, extend AM up to a carefully chosen point S such that BSCM becomes a parallelogram. (Hint: To arrange MC ∥ BS, first note that PM and MC are along the same line. Now where should M be on segment AS? Does your placement of S also prove MB ∥ CS?)

- Extend AM to S so that **M is the midpoint of AS** (MS = AM).
- In △ABS: P is the midpoint of AB and M is the midpoint of AS, so PM ∥ BS. PM and MC lie on one line, so **MC ∥ BS**.
- In △ACS: Q is the midpoint of AC and M is the midpoint of AS, so QM ∥ CS, that is, **MB ∥ CS**.
- So BSCM is a parallelogram, and its diagonals MS and BC bisect each other at X. So **X is the midpoint of BC**, and AX is the third median through M.
- Also MX = MS/2 = AM/2, so AM : MX = 2 : 1.

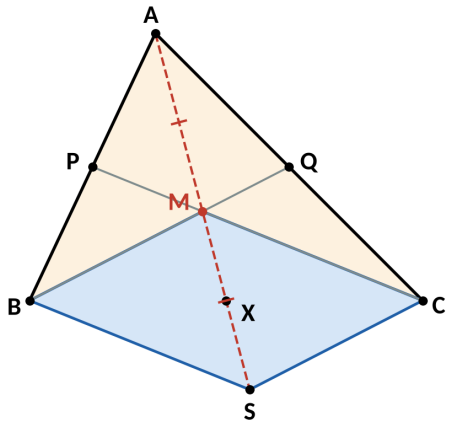


Fig. 17: BSCM is a parallelogram, so X is the midpoint of BC

Q20. In △ABC, let M and N be midpoints of AB and AC respectively. Let D be any point on BC. Show that MN bisects AD.

- Let MN meet AD at X. By the Midpoint Theorem, MN ∥ BC, so MX ∥ BD.
- In △ABD, M is the midpoint of AB and MX ∥ BD.
- By Theorem 7, X is the midpoint of AD. So **MN bisects AD**.

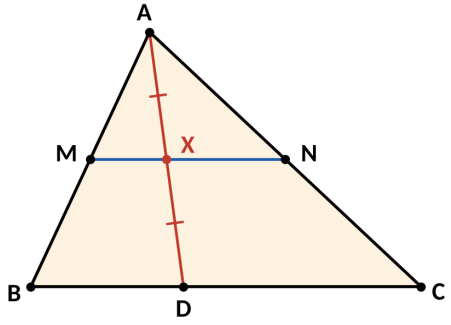


Fig. 18: MN bisects AD

Q21. In a quadrilateral ABCD, suppose AB ∥ DC and AB ≠ CD. Suppose G and H are the midpoints of AC and BD respectively. Prove that GH ∥ AB. (Why did we assume AB ≠ CD?)

- Let M be the midpoint of AD.
- In △ADC: M and G are midpoints of AD and AC, so MG ∥ DC ∥ AB.

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- In △DAB: M and H are midpoints of DA and DB, so MH ∥ AB.
- MG and MH are both parallel to AB through M, so they lie on one line.
- Hence **GH ∥ AB**.
- If AB = CD, then ABCD is a parallelogram (Theorem 5); its diagonals bisect each other, so G = H and there is no segment GH.

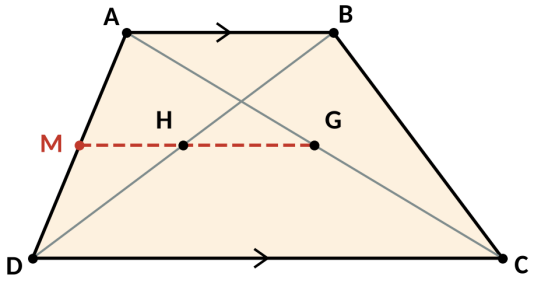


Fig. 19: G, H midpoints of the diagonals of trapezium ABCD

Q22. Suppose the midpoints of sides AB, BC, CD and DA of a quadrilateral ABCD are P, Q, R and S respectively. (i) Show that PR and QS bisect each other. (ii) Show that if AC = BD, then PR and QS are perpendicular. Is the converse true?

- (i) PQRS is a parallelogram (Theorem 9), and PR, QS are its diagonals.
- Diagonals of a parallelogram bisect each other.
- (ii) PQ = AC/2 and QR = BD/2.
- If AC = BD, then PQ = QR, so PQRS is a **rhombus**, and the diagonals of a rhombus are perpendicular.
- So PR ⊥ QS.
- **Converse is true:** if PR ⊥ QS, the parallelogram PQRS has perpendicular diagonals, so it is a rhombus (shown above)
- Then PQ = QR, so AC/2 = BD/2 and AC = BD.

Q23. Suppose PQRS is the Varignon parallelogram of ABCD. (i) Copy only PQRS on another paper. Show how you will recreate a congruent copy ABCD of ABCD from PQRS. This will be needed when we study tilings at the end of this chapter. (Hint: How will you place vertex A? How will you place B, C and D?) (ii) There are multiple ways to construct ABCD in (i). Justify why the quadrilateral ABCD you constructed is congruent to ABCD. You may need to show why S is collinear with the points A and D that you constructed, and similarly for P, Q and R. (Hint: Use congruence of triangles, for example △SDR ≅ △SDR.) (iii) Show that if PQRS is a square, then AC and BD are perpendicular and equal. Prove the converse.

- (i) Place A in the same position relative to S and P as A had: draw △SAP congruent to △SAP (SA = SA, PA = PA), on the side away from Q and R.
- Extend AP to B with PB = AP (P is the midpoint of AB)
- Extend BQ to C with QC = BQ. Extend CR to D with RD = CR. Join DA.
- (ii) Turning by 180° about P and then about Q moves every point by twice the vector PQ; turning about R and then about S moves it by twice SR = −PQ. Together, the four half-turns bring A back to itself.
- So S is the midpoint of DA: A, S, D are collinear.
- The original ABCD is produced from A by exactly the same four half-turns.
- A sits exactly as A did relative to PQRS, so the whole construction is a copy of the original: ABCD ≅ ABCD. (In particular △SDR ≅ △SDR, since SD = SA = SA = SD, RD = RC, and so on.)

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- (iii) PQ ∥ AC with PQ = AC/2, and QR ∥ BD with QR = BD/2.
- If PQRS is a square: PQ = QR gives **AC = BD**; PQ ⊥ QR gives **AC ⊥ BD**.
- **Converse:** if AC = BD and AC ⊥ BD, then PQ = QR and PQ ⊥ QR. A parallelogram with equal adjacent sides and a right angle is a square.

Q24. Mark the midpoint of the line drawn on the paper, given that the horizontal lines are equally spaced. Justify your answer.

- Count the gaps between ruled lines from one end of the drawn line to the other.
- If the number of gaps is even, the midpoint is where the drawn line crosses the **middle ruled line**. If it is odd, the midpoint is halfway between the two middle ruled lines along the drawn line.
- Justification: equally spaced parallel lines cut **equal parts** on any line crossing them.
- Take one gap: the parallel line midway between two ruled lines bisects every segment between them (Theorem 7 applied in the trapezium formed, or in a triangle)
- Repeating this, equal gaps give equal pieces of the drawn line.
- So the drawn line is cut into equal pieces, one per gap, and the middle of these pieces is the midpoint.

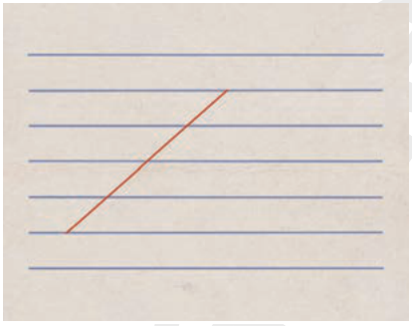


Fig. 20: A line drawn across equally spaced ruled lines

Q25. A right-triangle shaped cutout of a paper is folded such that point A touches point B. Show that the crease line can be used to find the midpoint of not only AB but also that of AC.

- When A is folded onto B, the crease is the perpendicular bisector of AB. It passes through the midpoint M of AB and is perpendicular to AB.
- The triangle is right-angled at B, so BC ⊥ AB. Two lines perpendicular to AB are parallel, so the crease BC.
- The crease passes through the midpoint M of AB and is parallel to BC. By Theorem 7, it meets AC at its **midpoint**.

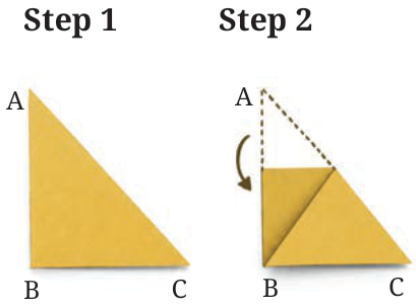


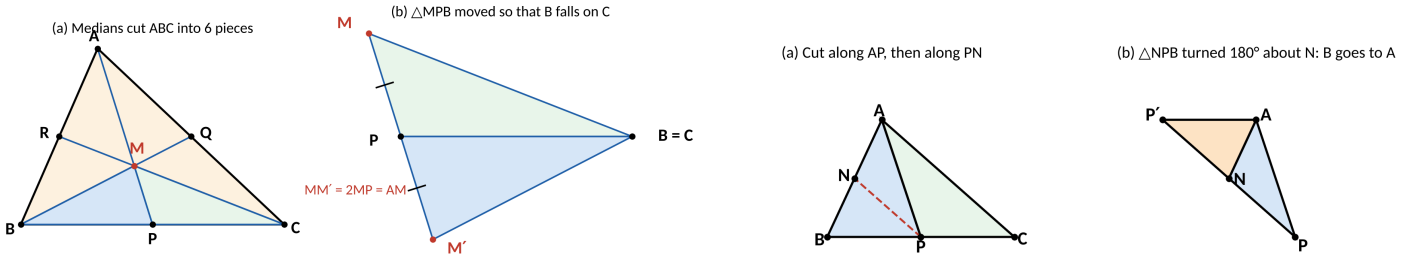
Fig. 21: Folding A onto B in a right triangle

Q26. You saw how to use the Midpoint Theorem to divide a given triangle into 4 congruent triangles. Can we divide a triangle into 3 congruent triangles? This question shows us how to do that and more, provided

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we are allowed to cut and reassemble. (i) Draw medians AP, BQ and CR of △ABC, meeting in a common point M. Cut along each median to get 6 triangles. Show with justification how to assemble the 6 pieces into 3 congruent triangles! (Hint: Align △MPB and △MPC along equal sides PB and PC, matching P with itself and B with C.) Prove that you get a triangle. What are its side lengths? (ii) Repeat the procedure with each of the 3 assembled triangles you got in (i). Show that each of the resulting 9 triangles has sides AB/3, BC/3 and AC/3. (iii) Can we cut a triangle and reassemble it into 2 congruent triangles? (Hint: Use median AP in △ABC and then a median of △APB.)

- (i) Move △MPB so that B falls on C and P stays at P; M goes to a new point M.
- ∠MPC + ∠MPC = ∠MPC + ∠MPB = 180° (linear pair)
- So M, P, M are collinear and the two pieces form **one triangle MMC**.
- Its sides: MC, MC = MB, and MM = 2MP = AM (centroid divides AP in 2 : 1)
- So the sides are **AM, BM, CM**, that is, ⅔ of each median.
- Doing the same with the pairs at Q and at R gives two more triangles with the same sides AM, BM, CM. So the 6 pieces make **3 congruent triangles**.
- (ii) A triangle whose sides are the medians of △ABC has medians equal to ¾BC, ¾CA, ¾AB. (Using the median length formula $m_a^2 = (2b^2 + 2c^2 - a^2)/4$ twice.)
- Each triangle in (i) has sides ⅔ of the medians, so its medians are ⅔ × ¾ = ½ of BC, CA, AB.
- Repeating (i) on it gives triangles with sides ⅔ of these medians = ⅔ × ½ = ⅓ of BC, CA, AB. So all 9 triangles have sides **AB/3, BC/3, AC/3**. (Lee Sallows found this in 2014.)
- (iii) **Yes.** Cut along median AP into △APB and △APC. Cut △APB along its median PN (N the midpoint of AB).
- Turn piece △NPB through 180° about N: B goes to A and P goes to P, with P, N, P collinear.
- The two pieces form △APP.
- AP = BP = PC, PP = 2PN = AC (Midpoint Theorem), AP common.
- So △APP ≅ △APC (SSS): **two congruent triangles**.



(i) pieces at P *(iii) two congruent triangles*

Fig. 22: Cutting and reassembling a triangle

- **Scissors congruence:** two figures with straight sides are scissors congruent if one can be cut with straight cuts and the pieces reassembled to make the other.

Q27. A more general midpoint theorem and its converse. In a quadrilateral ABCD, suppose AB ‖ DC. Recall that such ABCD is called a trapezium. Let E be the midpoint of AD. A line drawn through E intersects side BC at F. (i) If EF ‖ AB, then show that F is the midpoint of BC. Conclude that EF = (AB + CD)/2. (ii) If F is the midpoint of BC, then show that EF ‖ AB. There are at least two ways to solve (ii). Do it directly by using the midpoint M of BD and showing that EM and FM are the same lines. Or use (i) along with the same idea used in the second proof of the converse of the Midpoint Theorem. Which way do you think is simpler?

- (i) Let EF meet diagonal BD at M. In △DAB: E is the midpoint of AD and EM ‖ AB, so M is the midpoint of

BD and EM = AB/2 (Theorem 7).

- In △BDC: M is the midpoint of BD and MF ∥ DC, so F is the midpoint of BC and MF = DC/2.
- EF = EM + MF = **(AB + CD)/2**.
- (ii) **Direct way:** let M be the midpoint of BD. In △DAB, EM ∥ AB; in △BDC, MF ∥ DC ∥ AB. Both lines pass through M and are parallel to AB, so they are one line: E, M, F are collinear and **EF ∥ AB**.
- **Second way:** the line through E parallel to AB meets BC at its midpoint (by (i))
- That midpoint is F, and through E there is only one line to F, so EF is that parallel line.
- The second way is shorter once (i) is known.

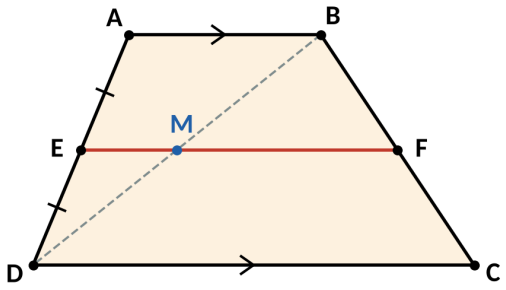


Fig. 23: Midpoints E, F of the non-parallel sides of a trapezium

Q28. ABCD is a trapezium with parallel sides AD = 3 cm and BC = 5 cm. E and F are the midpoints of the non-parallel sides. Find the ratio of the areas of the 4-gons AEFD and EBCF.

- EF ∥ AD ∥ BC and EF = (3 + 5)/2 = 4 cm (previous question).
- E is the midpoint of AB, so EF is halfway between AD and BC: if the height of ABCD is h, each part has height h/2.
- Area AEFD = ½ × (3 + 4) × h/2 = 7h/4.
- Area EBCF = ½ × (4 + 5) × h/2 = 9h/4.
- Ratio = **7 : 9**.

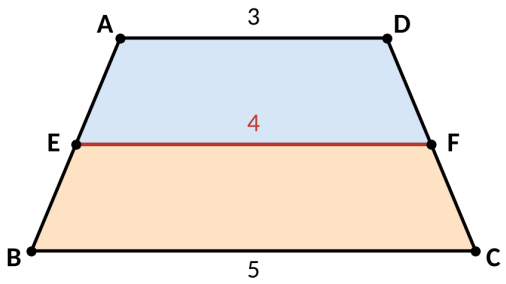


Fig. 24: Trapezium cut by EF into two parts

Q29. Suppose P is a point on side AB of △ABC and the line through P parallel to BC meets AC in point Q. For parts (i) to (iii), assume AP = 1, AQ = √5. (i) If PB = 2 find QC. (Hint: See the dotted line.) (ii) If PB = 1/3 find QC. (Hint: See the dotted lines.) (iii) If PB = 2/3 find QC. (Hint: Divide both PB and AP suitably.) (iv) Show that if AP/PB is a rational number, then AP/PB = AQ/QC. You will later prove this equality without assuming AP/PB is rational. That will require a new idea.

- Key fact: if equally spaced points on AB are joined by lines parallel to BC, these lines cut AC into **equal parts** (Theorem 7 and the trapezium result, used step by step).
- (i) Divide AB into 3 parts of length 1: AP is one part, PB two parts.
- Parallels through the division points cut AC into 3 equal parts, each equal to AQ = √5.

Conceptual Learning

- So **QC = 2√5**.
- (ii) Divide AB into parts of length 1/3: AP = 3 parts, PB = 1 part.
- AQ = √5 covers 3 parts, so each part is √5/3.
- **QC = √5/3**.
- (iii) Parts of length 1/3: AP = 3 parts, PB = 2 parts.
- Each part on AC is √5/3, so **QC = 2√5/3**.
- (iv) Let AP/PB = m/n.
- Take a unit u with AP = mu and PB = nu, so AB has m + n equal parts.
- Parallels to BC through the division points cut AC into m + n equal parts; AQ has m of them and QC has n.
- So AQ/QC = m/n = **AP/PB**.

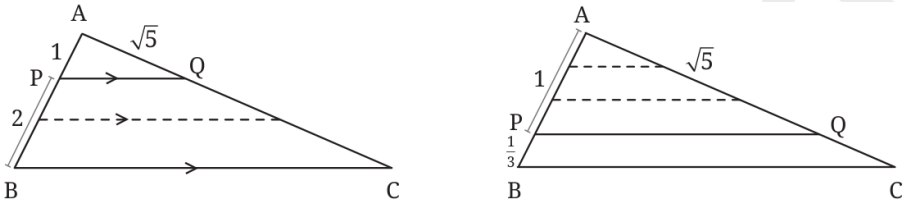


Fig. 25: Line PQ BC with AP = 1, AQ = √5

Q30. (i) Suppose ABCD is a parallelogram and M, N are midpoints of AB and CD respectively. Show that segments DM and BN trisect segment AC. (ii) Use part (i) to find a procedure to trisect any given segment PQ. Find two other ways to trisect PQ, one using the previous question and a third using the Centroid Theorem.

- (i) MB = AB/2 = CD/2 = DN and MB ∥ DN, so DMBN is a parallelogram (Theorem 5): DM ∥ BN.
- Let DM and BN meet AC at X and Y. In △ABY: M is the midpoint of AB and MX ∥ BY, so X is the midpoint of AY: AX = XY.
- In △CDX: N is the midpoint of CD and NY ∥ DX, so Y is the midpoint of CX: XY = YC. So **AX = XY = YC**.
- (ii) **Method 1:** take any point B not on PQ and complete the parallelogram PBQD with PQ as a diagonal.
- Mark midpoints M of PB and N of QD. DM and BN cut PQ at its trisection points.
- **Method 2:** draw a ray from P and mark three equal steps P_1, P_2, P_3. Join P_3Q and draw lines through P_1, P_2 parallel to P_3Q. They cut PQ into 3 equal parts.
- **Method 3:** take any point B, extend BQ to C with QC = BQ. Then PQ is a median of △PBC. Draw a second median (B to the midpoint of PC); it meets PQ at the centroid (the common point of the three medians) G with PG = ⅔PQ. G and the midpoint of PG trisect (divide into three equal parts) PQ.

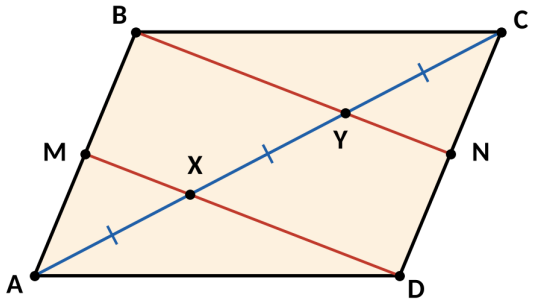


Fig. 26: DM and BN trisect AC

Q31. Is the Midpoint Theorem for Quadrilaterals (Theorem 9) true when the quadrilateral is non-convex? How about when it is self-intersecting? Experiment and check. The pictures may look strange, but the

theorem appears to be true. (i) Can you explain why this is so? (ii) There is one exception: In a very special case the Varignon parallelogram becomes a single segment. When will this happen? (iii) Does your proof apply even when ABCD is non-planar?

- (i) **Yes, it is true.** The proof uses only the Midpoint Theorem in △ABC, △ADC, △BCD, △BAD. These triangles exist for any four points with no three collinear, convex or not.
- So PQ ∥ AC ∥ SR and QR ∥ BD ∥ PS in every case.
- (ii) PQ ∥ AC and QR ∥ BD. If **AC ∥ BD**, then P, Q, R, S all lie on one line and the parallelogram flattens into a segment.
- This happens only for a self-intersecting ABCD (then A, B, D, C form a trapezium with parallel sides AC and BD).
- (iii) **Yes.** Each triangle is still planar, so PQ ∥ AC and SR ∥ AC with PQ = SR = AC/2.
- Two lines parallel to the same line are parallel, so P, Q, R, S lie in one plane and PQRS is a parallelogram even in space.

Q32. Let P, Q, R, S be four points on sides AB, BC, CD and DA respectively of a quadrilateral ABCD. Suppose PQRS is a parallelogram. Must P, Q, R and S be midpoints of the respective sides? This can be considered a possible converse question to Theorem 9.

- **No.** Example: let ABCD be a parallelogram with centre O.
- Take any P on AB and let R be its mirror point through O on CD (AP = CR)
- Take any Q on BC and its mirror point S on DA (BQ = DS).
- PR and QS both have midpoint O, so PQRS is a parallelogram (Theorem 4), but P, Q, R, S need not be midpoints.

Q33. (i) Complete the following proof of the Midpoint Theorem. Extend segment PQ beyond Q until point S. By how much should we extend PQ? It would be good to be able to prove that △APQ ≅ △CSQ. (Why?) Use this as a guide to specify the location of S and then complete this proof. (ii) Give a similar proof of the converse of the Midpoint Theorem (Theorem 7). Start by extending PQ up to a suitable point S.

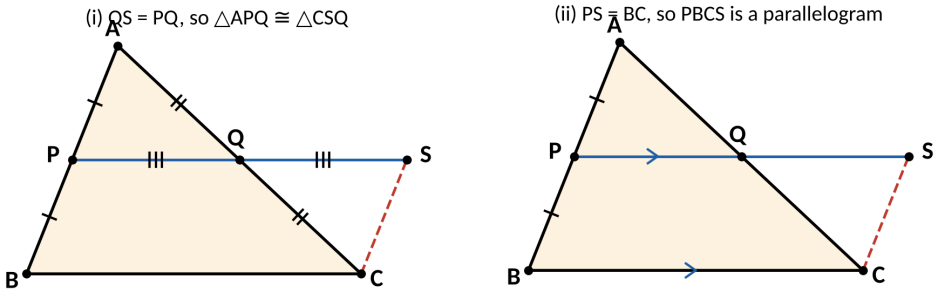


Fig. 27: Proof by extending PQ to S

- (i) Extend PQ to S with **QS = PQ**. Then △APQ ≅ △CSQ (SAS: AQ = CQ, PQ = SQ, vertically opposite angles at Q).
- We want this congruence because it moves AP to CS: CS = AP = PB, and ∠QCS = ∠QAP gives CS ∥ AB, that is, CS ∥ PB.
- CS = PB and CS ∥ PB, so PBCS is a parallelogram (Theorem 5)
- Hence PS ∥ BC and PS = BC, so **PQ ∥ BC** and **PQ = PS/2 = BC/2**.
- (ii) P is the midpoint of AB and PQ ∥ BC. Extend PQ to S with **PS = BC**. Then PS ∥ BC and PS = BC, so PBCS is a parallelogram: CS = PB = AP and CS ∥ AB.
- △APQ ≅ △CSQ (AAS: ∠APQ = ∠CSQ and ∠PAQ = ∠SCQ as alternate angles, AP = CS)

Conceptual Learning

➢ So **AQ = QC** and PQ = QS = BC/2.

Q34. Multiple converses to a theorem. The Midpoint Theorem starts with △ABC and points P and Q on sides AB and AC. Assumptions: (P MID) P is the midpoint of AB, and (Q MID) Q is the midpoint of AC. Conclusions: (PRLL) PQ ‖ BC, and (HALF) PQ = BC/2. (i) A natural converse of the Midpoint Theorem would be: "If (PRLL) and (HALF) are true, then (P MID) and (Q MID) are true." Write out this statement fully. Experiment and see that it seems to be true! Prove the statement. (Hint: Try to run a proof of the Midpoint Theorem backwards.) (ii) To see Theorem 7 as a converse of the Midpoint Theorem, we first write the Midpoint Theorem as follows: "Suppose P is the midpoint of side AB of △ABC and Q is a point on side AC. If Q is the midpoint of AC then PQ ‖ BC." Write the converse of the second sentence in quotes by keeping the first sentence the same. Verify that Theorem 7 is what you get! Now write Theorem 7 in terms of the four named conditions. (iii) It is sensible to take two of the four listed statements as assumptions and ask if the other two are true. Verify that only one such combination remains to be examined. "If (P MID) and (HALF) are true, then can we conclude (Q MID) and/or (PRLL)?" Write this out in words. Can you answer it? Why did we focus only on Theorem 7 among the multiple possibilities?

➢ (i) Statement: "In △ABC, P is on AB and Q is on AC. If PQ ‖ BC and PQ = BC/2, then P and Q are the midpoints of AB and AC."

➢ Proof: draw QR ‖ AB meeting BC at R. PBRQ is a parallelogram, so BR = PQ = BC/2: R is the midpoint of BC, and RQ = PB.

➢ In △CBA, R is the midpoint of BC and RQ ‖ BA, so Q is the midpoint of AC and RQ = AB/2 (Theorem 7)

➢ Then PB = RQ = AB/2, so P is the midpoint of AB. **True.**

➢ (ii) Converse of the second sentence: "If PQ ‖ BC, then Q is the midpoint of AC." With the first sentence kept, this is exactly **Theorem 7**. In named conditions: **if (P MID) and (PRLL), then (Q MID) and (HALF)**.

➢ (iii) Pairs of assumptions: {P MID, Q MID} is the Midpoint Theorem; {PRLL, HALF} is (i); {P MID, PRLL} is Theorem 7, and {Q MID, PRLL} is Theorem 7 with the roles of AB and AC swapped; {P MID, HALF} and {Q MID, HALF} are mirror cases.

➢ So **only (P MID) + (HALF) remains**.

➢ In words: "P is the midpoint of AB and Q is a point on AC with PQ = BC/2.

➢ Must Q be the midpoint of AC, or PQ be parallel to BC?"

➢ Answer: **not necessarily.** The circle with centre P and radius BC/2 passes through the midpoint M of AC, but it can meet AC at a second point Q as well.

➢ For that Q, PQ = BC/2 but Q is not the midpoint and PQ is not parallel to BC.

➢ Theorem 7 was chosen because it is the most useful of these statements.

Q35. What fraction of the square is shaded? (Each side of the square is bisected; each corner is joined to the midpoint of a side as shown.)

➢ Take the square of side 2 with corners (0, 0), (2, 0), (2, 2), (0, 2).

➢ Two of the drawn lines are 2x + y = 2 (from (0, 2) to (1, 0)) and 2x + y = 4 (from (1, 2) to (2, 0))

➢ They are parallel; distance between them = $2/\sqrt{5}$.

➢ The other two lines are also parallel and $2/\sqrt{5}$ apart, and are perpendicular to the first pair.

➢ So the shaded figure is a square of side $2/\sqrt{5}$.

➢ Shaded area = 4/5; square area = 4.

➢ Fraction = **1/5**.

Conceptual Learnings

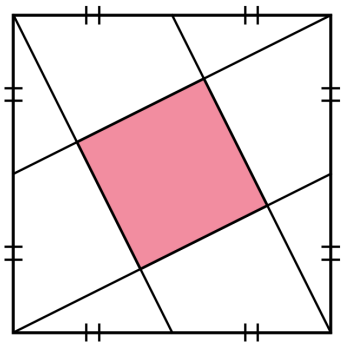


Fig. 28: Shaded square inside a square

Tiling the Plane Using Any 4-gon

✓ **Tilings around us:**

- Tilings appear in ancient mosaics (pictures made from small pieces fitted together), in artwork such as that of M. C. Escher, in nature (a honeycomb), and in everyday decorative patterns.

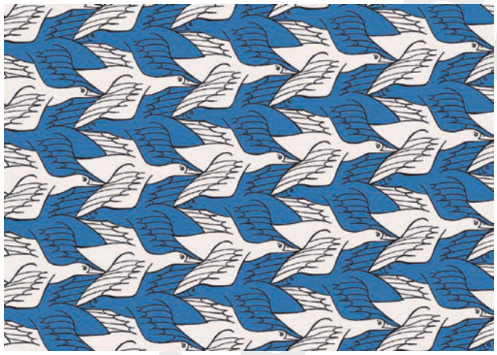
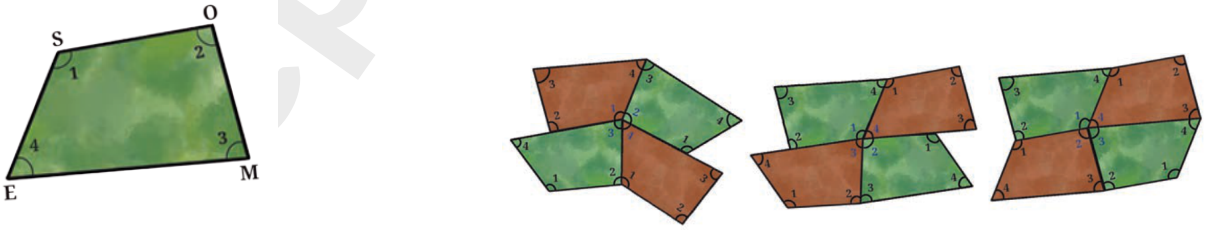


Fig. 29: A tiling pattern with bird-shaped tiles

✓ **Angles at a vertex:**

- In any tiling, the angles that meet at a vertex add up to **360°**.
- The four angles of a quadrilateral also add up to 360°. So four copies of any 4-gon SOME can be fitted around one point, using each angle (1, 2, 3, 4) once.



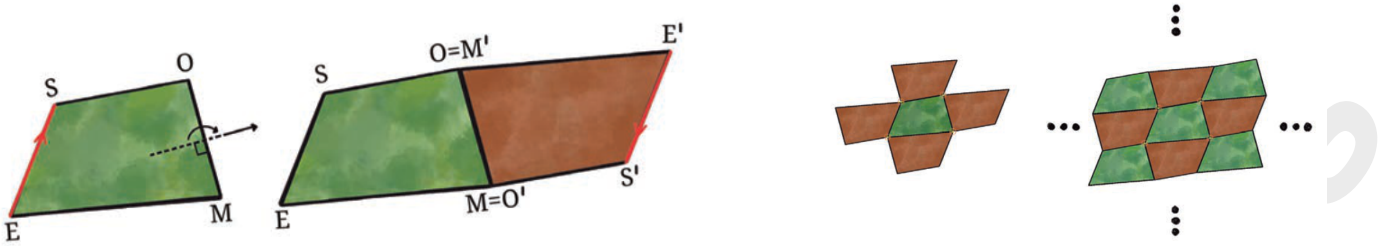
(a) 4-gon SOME *(b) Four copies around a point*

Fig. 30: Four copies of SOME around a point, each angle used once

✓ **Method 1: Half-turns about midpoints of sides:**

1) Start with one copy of SOME.
2) Turn a second copy by 180° about the midpoint of side OM. It fits exactly along the shared side OM.
3) Repeat this at every side of every copy that is added.
4) Each new copy can be obtained by turning either of its neighbours, and both ways give the same

result; the new copies fit without gaps or overlaps. This builds a tiling of the whole plane.



(a) Half-turn about the midpoint of OM *(b) Two steps of adding copies*

Fig. 31: Method 1

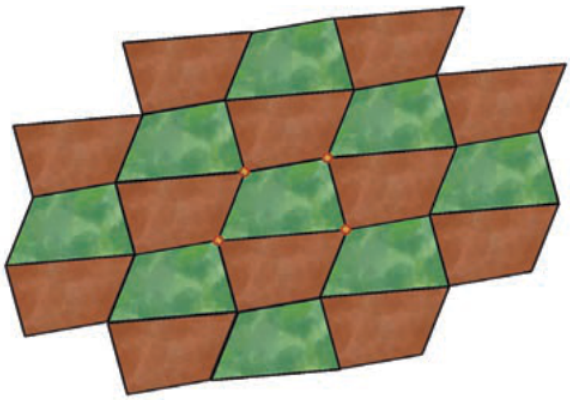


Fig. 32: Tiling of the plane by copies of SOME

✓ **Method 2: Using the Varignon grid:**

- Draw the Varignon parallelogram in every copy of SOME in the tiling: these parallelograms form a **grid**, the same parallelogram tiling as the hinged sticks.
- So start with the grid of Varignon parallelograms of SOME. Place copies of SOME on alternate grid cells (like the black squares of a chess board), matching each copy to its Varignon parallelogram.
- The placed copies meet only at their vertices. The gaps left between them are automatically copies of SOME. Filling them completes the tiling.



(a) Varignon parallelograms in the tiling *(b) Grid with alternate cells shaded*

Fig. 33: Method 2

✓ **Aperiodic tilings and quasicrystals:**

- **Aperiodic** (never repeating in a regular pattern) **tiling:** a tiling that covers the plane endlessly but never repeats in a regular pattern.
- Such tilings were found useful for describing the atoms of **quasicrystals** (solids whose atoms are ordered but never repeat in a regular pattern), discovered by Dan Shechtman. He won the Nobel

Prize in Chemistry in 2011.

✓ **The hat tile (2023):**

- In 2023, Smith, Myers, Kaplan and Goodman-Strauss found the first single shape, called the **'hat'**, that tiles the plane only aperiodically: it covers the plane endlessly without ever repeating the same pattern.

✓ **Solved questions:**

Q36. Using a fact about parallelograms, show how to tile the plane using any given triangle. (Hint: Can you use the parallelogram tiling in the introduction?)

- Take △ABC. Turn a copy by 180° about the midpoint of BC. The two copies form quadrilateral ABDC.
- Its diagonals AD and BC bisect each other, so it is a **parallelogram** (Theorem 4).
- Parallelograms tile the plane (hinged grid)
- Each parallelogram is two copies of the triangle, so the **triangle tiles the plane**.

Q37. Justify why the plane cannot be tiled with a regular pentagon. (Hint: think about the angles that meet at a vertex of the tiling.)

- Each angle of a regular pentagon = (5 − 2) × 180° ÷ 5 = **108°**.
- At a vertex of the tiling, the angles must add up to 360°.
- With only corners meeting: 108k = 360 gives k = 3.33, not a whole number (3 corners leave a 36° gap; 4 corners overlap).
- If a corner touches the middle of another tile's side, the angles must add up to 180° there: 108k = 180 gives k = 1.67, again impossible.
- So regular pentagons cannot tile the plane.
- (Some **irregular** pentagons can; the 15th such type was found in 2015.)



(a) Three regular pentagons leave a gap *(b) Tiling by an irregular pentagon*

Fig. 34: Regular and irregular pentagons

Q38. Draw a non-convex 4-gon DART. Show how we can tile the plane with copies of DART. Both methods that we discussed will work. Which do you prefer?

- Draw DART and turn a copy by 180° about the midpoint of each side (Method 1)
- Keep repeating for every new copy.
- The angles at every vertex are the four angles of DART, one each, adding up to 360° (the reflex angle at the dent fills more than half a turn).
- Method 2 also works: draw the Varignon parallelogram of DART (Theorem 9 holds for non-convex 4-gons), make its grid and place copies on alternate cells.

Conceptual Learning

- Method 1 is simpler to draw by hand; Method 2 shows clearly why the pattern repeats.

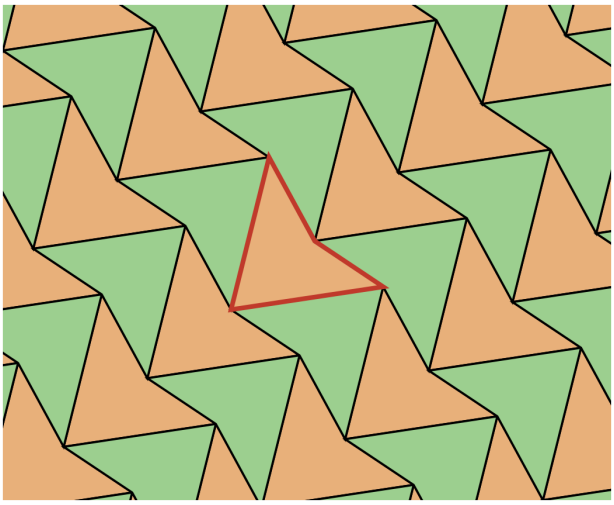


Fig. 35: Tiling the plane with a non-convex 4-gon (one copy outlined in red)

Q39. Let us see a third method to tile the plane using a 4-gon. Focus on only two coloured copies of SOME sharing a vertex: each 4-gon is just a shifted copy of the other. Draw two copies of SOME so that the diagonals EO and EO are collinear with O = E. Place a cutout of SOME (with diagonal EO drawn) on top of SOME and slide it so that segment EO moves along EO until EO matches EO. Verify that your cutout exactly matches SOME. (i) Prove the exact match of SOME with SOME using four parallelograms. (ii) Follow the described procedure along each diagonal of the starting coloured 4-gon in both directions to get 4 new 4-gons. Using these 4 copies, repeat the procedure 4 more times to place 4 new copies, resulting in a 3 by 3 grid of 9 copies. These 9 copies meet each other only at vertices and once again the blank spaces among them also form 4-gons congruent to SOME!

- (i) Sliding along the diagonal moves every point by the same distance EO in the same direction (a **translation**).
- So SS, OO, MM, EE are all equal to EO and parallel to EO. Then SSOO, OOMM, MMEE and EESS each have one pair of opposite sides equal and parallel, so they are **parallelograms** (Theorem 5).
- Opposite sides of a parallelogram are equal, so SO = SO, OM = OM, ME = ME, ES = ES, and the angles also match.
- Hence SOME is an exact copy of SOME.
- (ii) Sliding along diagonal EO (both ways) and along diagonal SM (both ways) gives 4 new copies; sliding these again gives the 3 by 3 grid of 9 copies.
- The copies touch only at vertices, and each gap between four of them is bounded by four sides of different copies: one of each length, with angles that fill it exactly.
- So each gap is another copy of SOME (turned through 180°), and the plane is covered.

Q40. Validity of tiling methods. Show by proof that each of the three tiling methods we saw produces a tiling of the plane using the given 4-gon. You have to prove that when we place new copies using any of the three procedures, the entire plane is covered with no gaps and no overlaps. First, think carefully about what it means to prove this. Then find a proof for each procedure.

- **What must be shown:** (1) every point of the plane lies in some copy (no gaps), and (2) no two copies overlap except along sides.
- Let the 4-gon have vertices A, B, C, D. Half-turns about the midpoints of AB and BC together give a slide by vector AC; half-turns about the midpoints of BC and CD give a slide by vector BD.

Conceptual Learning

- So all copies are of two kinds: the 4-gon slid by whole-number combinations of AC and BD, and its half-turned copy slid in the same way.
- These are exactly the copies in Method 1 and Method 3 (Method 3 slides along the diagonals).
- Each original copy together with one half-turned neighbour forms a region whose Varignon parallelogram grid cell (sides AC/2 and BD/2, area half the parallelogram on AC and BD) repeats by these slides.
- The grid of parallelograms covers the plane with no gaps or overlaps, so the copies built on it (Method 2) do too.
- At every vertex, the four angles of the 4-gon meet once each, totalling 360°, and each side is shared by exactly two copies.
- So no gaps or overlaps arise anywhere, and all three methods give a tiling.

Angles, Sides and Diagonals of Polygons

✓ **Solved questions:**

Q41. You know that the sum of angles of a quadrilateral is 360°, even for a non-convex quadrilateral. (Recall the proof.) Now consider a self-intersecting quadrilateral ABCD, where AB and CD intersect at point E. Show that ∠A + ∠B + ∠C + ∠D < 360°. Can you construct ABCD such that ∠A + ∠B + ∠C + ∠D = 2°?

- Proof for an ordinary quadrilateral: a diagonal splits it into two triangles, so the angle sum is 2 × 180° = 360°.
- Self-intersecting ABCD: let ∠AED = ∠BEC = θ (vertically opposite).
- In △AED: ∠A + ∠D = 180° − θ. In △BEC: ∠B + ∠C = 180° − θ.
- So ∠A + ∠B + ∠C + ∠D = 360° − 2θ, which is **less than 360°**.
- **Yes:** for a sum of 2°, make 360° − 2θ = 2°, that is, θ = 179°.
- Draw two lines crossing at E at 179°, and place A, B on one line and D, C on the other on opposite sides of E.

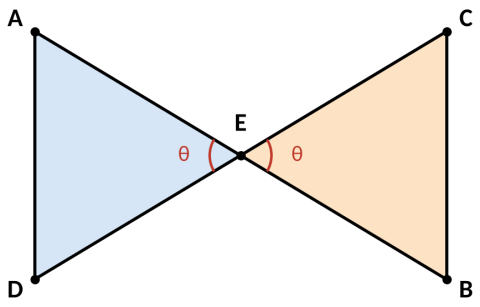


Fig. 36: Self-intersecting ABCD with AB and CD crossing at E

Q42. Consider 4 points A, B, C, D in the plane with no three collinear. Answer the following questions. Some answers may require you to consider different cases, depending on how the points are positioned in the plane. (i) How many different quadrilaterals do they form if the quadrilateral is allowed to be self-intersecting or non-convex? (ii) How many of these quadrilaterals are self-intersecting? How many are convex?

- (i) Each quadrilateral is an order of going round the 4 points. 4! = 24 orders, but each quadrilateral has 8 names (4 starting points × 2 directions)
- So there are 24 ÷ 8 = **3 quadrilaterals**: ABCD, ABDC, ACBD.

- (ii) **Case 1: the four points form a convex shape** (none inside the triangle of the others): **1 convex** and **2 self-intersecting**.
- **Case 2: one point lies inside the triangle formed by the other three: 0 convex**, **0 self-intersecting**; all 3 are non-convex.

Q43. Show that the sum of angles of a non-planar quadrilateral is always less than 360°. Can you find a non-planar quadrilateral ABCD for which ∠A + ∠B + ∠C + ∠D = 2°? (Hint: Think of a diagonal, say AC, as a hinge around which triangles ABC and ADC can rotate.) What happens to each angle of ABCD as you do this rotation?

- Diagonal AC splits ABCD into △ABC and △ADC, whose angles add up to 360°.
- At A, the angle ∠DAB is less than ∠DAC + ∠CAB when D, A, B, C are not in one plane (the angle between two rays is the shortest turn, and here the path via AC bends out of the plane)
- Likewise ∠BCD < ∠BCA + ∠ACD.
- ∠B and ∠D are the same as in the triangles.
- So ∠A + ∠B + ∠C + ∠D < 360°.
- As △ADC turns about the hinge (a joint about which something turns) AC, ∠B and ∠D stay fixed while ∠A and ∠C shrink, reaching |∠DAC − ∠CAB| and |∠ACD − ∠BCA| when the 4-gon is folded flat.
- **Yes:** take two congruent thin triangles ABC and ADC with ∠B = ∠D = 0.5° and fold them almost shut about AC, so that ∠A and ∠C are 0.5° each.
- Then the sum is about **2°**.

Q44. Is there a 4-gon with given side lengths? (i) For given positive numbers a, b, c, is there a triangle whose sides have these lengths? The answer is Yes exactly when the sum of any two numbers is greater than the third. If a ≤ b ≤ c, this amounts to requiring a + b > c. Check this fact by construction. (Hint: Start by drawing a segment of length c.) (ii) Suppose a 4-gon has 2, 5, 11 as three side lengths. Can the length of the fourth side be 100? Can it be 10? Can it be 1? What are the possible lengths of the fourth side? (iii) For given positive numbers a, b, c, d, how will you decide if there is a 4-gon whose sides have these lengths?

- (i) Draw a segment of length c.
- Draw circles of radii a and b about its ends.
- They meet (giving the third vertex) exactly when a + b > c.
- Example: 3, 4, 8 fails (3 + 4 < 8); 3, 4, 6 works.
- (ii) A 4-gon exists only if the longest side is less than the sum of the other three.
- **100: No** (2 + 5 + 11 = 18 < 100)
- **10: Yes** (longest 11 < 2 + 5 + 10 = 17)
- **1: No** (longest 11 is not less than 2 + 5 + 1 = 8).
- Possible fourth side x: x < 2 + 5 + 11 = 18 and 11 < 2 + 5 + x, so **4 < x < 18**.
- (iii) A 4-gon with sides a, b, c, d exists exactly when **each length is less than the sum of the other three** (equivalently, the longest is less than the sum of the rest).

Q45. Counting diagonals of a polygon. (i) How should we describe a diagonal of an n-gon? How many diagonals does an n-gon have? Make a table for small values of n. A 3-gon has no diagonals. A 4-gon has 2. How many diagonals does a 5-gon have? A 6-gon? Try to find a pattern and guess the answers for n = 7 and n = 8. Check your guesses by systematic counting. (ii) Can you guess a formula for the number of diagonals? How many diagonals get added when we increase the number of sides by 1? Can you now justify why the formula you guessed is true for all n?

Conceptual Learning

➢ (i) **Diagonal:** a segment joining two vertices that are not adjacent.

n	3	4	5	6	7	8
Diagonals	0	2	5	9	14	20
Added from previous	–	2	3	4	5	6

➢ (ii) Formula: number of diagonals = **n(n − 3)/2**.

➢ Proof: each vertex joins to n − 3 others by diagonals (not itself and not its 2 neighbours). n vertices give n(n − 3), but each diagonal is counted twice, so divide by 2.

➢ Going from n to n + 1 sides adds **n − 1** diagonals: the new vertex gives n − 2, and the old side joining its two neighbours becomes 1 more diagonal.

➢ Check: n(n − 3)/2 + (n − 1) = (n + 1)(n − 2)/2 ✓.

Q46. Sum of angles of a polygon. What is the sum of angles of a (planar non-self-intersecting) n-gon? We know that the answer is 180° for n = 3 and 360° for n = 4. Find the next few values. Then find a formula in terms of n and prove it.

➢ 5-gon: 540°; 6-gon: 720°; 7-gon: 900°; 8-gon: 1080°.

➢ Formula: sum of angles = **(n − 2) × 180°**.

➢ Proof (convex n-gon): diagonals from one vertex split it into n − 2 triangles, and their angles together make up the angles of the n-gon.

➢ So the sum is (n − 2) × 180°.

➢ Every planar non-self-intersecting n-gon (convex or not) can be cut by suitable diagonals into n − 2 triangles, so the same formula holds.

Activities

✓ **Activity 1: Hinged grid of sticks:**

Aim:

- To see that copies of any parallelogram tile the plane.

Steps:

1) Make a grid of rectangles with sticks (or straws), pinning them at each crossing so they can turn.
2) Hold the bottom stick fixed and push a vertical side sideways.

Observations:

- Each set of sticks stays parallel; the rectangles become equal parallelograms with no gaps or overlaps.

Conclusion:

- The plane can be tiled with copies of any parallelogram.

✓ **Activity 2: Cutting a triangle at the midpoints:**

Aim:

- To find the relation between the four triangles formed by joining the midpoints of the sides of a triangle.

Steps:

1) Cut out two copies of the same triangle. Keep one aside.

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2) On the other, mark the midpoint of each side by folding (bring two vertices together).
3) Join the three midpoints and cut along these lines to get four small triangles.
4) Compare the four triangles with each other and with the intact triangle. Repeat for triangles of other shapes.

- **Observations:**
  - All four small triangles fit exactly on each other. Each looks like the original triangle at half the size.
  - Each midpoint segment is parallel to the third side and half its length.
- **Conclusion:**
  - The midpoints cut a triangle into four congruent triangles; this leads to the Midpoint Theorem.

✓ **Activity 3: Medians of a triangle:**

- **Aim:**
  - To check whether the three medians of a triangle meet at one point, and how they are divided.
- **Steps:**
  1) Draw a triangle ABC and mark the midpoints P, Q, R of AB, AC, BC.
  2) Draw the medians AR, BQ and CP.
  3) Measure the two parts of each median at the common point M. Repeat for several triangles.
- **Observations:**
  - The three medians always pass through one point M.
  - On every median, the part from the vertex to M is twice the part from M to the midpoint (e.g. CM = 4 cm, MP = 2 cm).
- **Conclusion:**
  - The medians are concurrent at the centroid, which divides each median in the ratio 2 : 1.

✓ **Activity 4: Tiling with an irregular 4-gon:**

- **Aim:**
  - To tile the plane with copies of an irregular quadrilateral SOME.
- **Steps:**
  1) Trace SOME on paper and label its angles 1, 2, 3, 4. Cut out 15 identical copies.
  2) Fit 4 copies around one point so that each angle is used once.
  3) Continue by turning copies through 180° about the midpoints of their sides, and fit more pieces.
- **Observations:**
  - At every vertex, the angles 1, 2, 3, 4 meet once each and fill 360°.
  - The copies fit without gaps or overlaps, and a repeating pattern appears.
- **Conclusion:**
  - Any quadrilateral, even an irregular one, can tile the plane.

✓ **Activity 5: The hat tile:**

- **Aim:**
  - To explore a tiling that never repeats.

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- **Steps:**
  1) Cut out 15 identical copies of the ‘hat’ shape.
  2) Fit them together edge to edge to cover a region of the table.
- **Observations:**
  - The hats fit together without gaps, but no fixed pattern keeps repeating.
- **Conclusion:**
  - The hat tiles the plane only **aperiodically**: endlessly, but without ever repeating the same pattern.

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